REFUTATION OF THE LINEAR-SAVING CONJECTURE FOR TRIANGLE-FREE DIAMETER-FOUR COMPLETION (ERDŐS PROBLEM #619)

ARGUS OPEN RESEARCH PROJECT

Abstract. For a connected triangle-free graph G on n vertices, let $h_4(G)$ denote the minimum number of missing edges that can be added to G on the same vertex set so that the resulting supergraph $H \supseteq G$ remains triangle-free and has diameter at most 4. Problem 4.3 of Erdős, Gyárfás, and Ruszinkó (1998; Erdős Problem #619) asks whether there exists a constant $\epsilon > 0$ such that $h_4(G) \le (1-\epsilon)n$ for every connected triangle-free graph G on n vertices. We document and verify the Lean 4 disproof showing that for every $\eta \in (0,1)$ and all sufficiently large n, there exist connected triangle-free graphs G on n vertices with $h_4(G) \ge (1-\eta)n$.

1. Main Theorem

Theorem 1.1 (Formally Verified in Lean 4)**.** *For every real $\eta \in (0,1)$, there exists $N_0(\eta)$ such that for every $n \ge N_0(\eta)$ there is a connected triangle-free graph G on n vertices (constructed by attaching pendant vertices to a dense triangle-free core of independence number $o(n)$) such that every triangle-free supergraph $H \supseteq G$ on $V(G)$ with* $\operatorname{diam}(H) \le 4$ *satisfies*

$$|E(H) \setminus E(G)| \;\ge\; (1-\eta)n,$$

and therefore $h_4(G) \ge (1-\eta)n$, refuting the existence of a uniform constant $\epsilon > 0$ with $h_4(G) \le (1-\epsilon)n$.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2215 in the domain of extremal graph theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *formal artifact refutation* deciding the scope: connected triangle-free graphs on n vertices requiring h_4(G) >= (1-eta)n new edges for every eta > 0 (Lean 4 verified).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- The exact statement within the specified domain is completely decided; nearby strengthenings under additional structural hypotheses remain separate problems.

References

[1] Primary problem list and source literature for Result 2215 (extremal graph theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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